Table1 EMPLOYEES TABLE

```sql
-- Employees Table
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    dept_id INT,
    salary INT,
    hire_date DATE,
    city VARCHAR(50)
);

INSERT INTO employees VALUES
(101, 'Anuj', 1, 50000, '2022-01-10', 'Delhi'),
(102, 'Ravi', 2, 60000, '2021-03-15', 'Mumbai'),
(103, 'Neha', 1, 55000, '2020-07-20', 'Delhi'),
(104, 'Amit', 3, 70000, '2019-11-01', 'Pune'),
(105, 'Sneha', 2, 65000, '2023-05-12', 'Mumbai'),
(106, 'Karan', 3, 72000, '2018-08-25', 'Bangalore'),
(107, 'Pooja', 1, 48000, '2022-09-30', 'Delhi'),
(108, 'Raj', 2, 52000, '2021-12-11', 'Chennai'),
(109, 'Meena', 3, 68000, '2020-02-14', 'Pune'),
(110, 'Vikas', 1, 51000, '2023-01-18', 'Delhi');
```

```
mysql> SELECT * FROM employees;

+--------+----------+---------+--------+------------+-----------+
| emp_id | emp_name | dept_id| salary | hire_date  | city      |
+--------+----------+---------+--------+------------+-----------+
|    101 | Anuj     |       1 |  50000 | 2022-01-10 | Delhi     |
|    102 | Ravi     |       2 |  60000 | 2021-03-15 | Mumbai    |
|    103 | Neha     |       1 |  55000 | 2020-07-20 | Delhi     |
|    104 | Amit     |       3 |  70000 | 2019-11-01 | Pune      |
|    105 | Sneha    |       2 |  65000 | 2023-05-12 | Mumbai    |
|    106 | Karan    |       3 |  72000 | 2018-08-25 | Bangalore |
|    107 | Pooja    |       1 |  48000 | 2022-09-30 | Delhi     |
|    108 | Raj      |       2 |  52000 | 2021-12-11 | Chennai   |
|    109 | Meena    |       3 |  68000 | 2020-02-14 | Pune      |
|    110 | Vikas    |       1 |  51000 | 2023-01-18 | Delhi     |
+--------+----------+---------+--------+------------+-----------+
10 rows in set (0.00 sec)
```

Table 2 DEPATEMENTS

```sql
-- Departments Table
DROP TABLE IF EXISTS departments;
CREATE TABLE departments (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50)
);

INSERT INTO departments VALUES
(1, 'HR'),
(2, 'Sales'),
(3, 'IT');
```

```
mysql> SELECT * FROM departments;

+---------+-----------+
| dept_id | dept_name |
+---------+-----------+
|       1 | HR        |
|       2 | Sales     |
|       3 | IT        |
+---------+-----------+
3 rows in set (0.00 sec)
```

Table3 PROJECTS

```sql
-- Projects Table
DROP TABLE IF EXISTS projects;
CREATE TABLE projects (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(50),
    dept_id INT,
    budget INT
);

INSERT INTO projects VALUES
(201, 'Alpha', 1, 100000),
(202, 'Beta', 2, 150000),
(203, 'Gamma', 3, 200000),
(204, 'Delta', 2, 120000);
```

```
mysql> SELECT * FROM projects;

+------------+--------------+---------+--------+
| project_id | project_name | dept_id | budget |
+------------+--------------+---------+--------+
|        201 | Alpha        |       1 | 100000 |
|        202 | Beta         |       2 | 150000 |
|        203 | Gamma        |       3 | 200000 |
|        204 | Delta        |       2 | 120000 |
+------------+--------------+---------+--------+
4 rows in set (0.00 sec)
```

Table4 ATTENDANCE

```sql
-- Attendance Table
DROP TABLE IF EXISTS attendance;
CREATE TABLE attendance (
    emp_id INT,
    date DATE,
    status VARCHAR(20)
);

INSERT INTO attendance VALUES
(101, '2024-03-01', 'Present'),
(102, '2024-03-01', 'Absent'),
(103, '2024-03-01', 'Present'),
(104, '2024-03-01', 'Present'),
(105, '2024-03-01', 'Absent'),
(106, '2024-03-01', 'Present'),
(107, '2024-03-01', 'Present'),
(108, '2024-03-01', 'Present'),
(109, '2024-03-01', 'Absent'),
(110, '2024-03-01', 'Present');
```

```
mysql> SELECT * FROM attendance;

+--------+------------+---------+
| emp_id | date       | status  |
+--------+------------+---------+
|    101 | 2024-03-01 | Present |
|    102 | 2024-03-01 | Absent  |
|    103 | 2024-03-01 | Present |
|    104 | 2024-03-01 | Present |
|    105 | 2024-03-01 | Absent  |
|    106 | 2024-03-01 | Present |
|    107 | 2024-03-01 | Present |
|    108 | 2024-03-01 | Present |
|    109 | 2024-03-01 | Absent  |
|    110 | 2024-03-01 | Present |
+--------+------------+---------+
10 rows in set (0.00 sec)
```

SQL Questions (Lab Practice)

Basic Level (1–10)

1. Retrieve all employees

```
SELECT * FROM employees;
```

```
+--------+----------+---------+--------+------------+----------+
| emp_id | emp_name | dept_id | salary | hire_date  | city     |
+--------+----------+---------+--------+------------+----------+
| 101    | Ravi     |       1 |  50000 | 2022-01-10 | Delhi    |
| 102    | Ravi     |       2 |  60000 | 2021-03-15 | Mumbai   |
| 103    | Neha     |       1 |  55000 | 2020-07-20 | Delhi    |
| 104    | Amit     |       3 |  70000 | 2019-11-01 | Pune     |
| 105    | Sneha    |       2 |  65000 | 2023-05-12 | Mumbai   |
| 106    | Karan    |       3 |  72000 | 2018-08-25 | Bangalore |
| 107    | Pooja    |       1 |  48000 | 2022-09-30 | Delhi    |
| 108    | Raj      |       2 |  52000 | 2021-12-11 | Chennai  |
| 109    | Meena    |       3 |  68000 | 2020-02-14 | Pune     |
| 110    | Vikas    |       1 |  51000 | 2023-01-18 | Delhi    |
+--------+----------+---------+--------+------------+----------+
10 rows in set (0.00 sec)
```

2. Display only emp_name and salary.

```
SELECT emp name, salary FROM employees;
```

```
+----------+--------+
| emp_name | salary |
+----------+--------+
| Anuj     |  50000 |
| Ravi     |  60000 |
| Neha     |  55000 |
| Amit     |  70000 |
| Sneha    |  65000 |
| Karan    |  72000 |
| Pooja    |  48000 |
| Raj      |  52000 |
| Meena    |  68000 |
| Vikas    |  51000 |
+----------+--------+
10 rows in set (0.00 sec)
```

3. Find employees working in department 1.

```
SELECT * FROM employees WHERE dept id = 1;
```

```
+--------+----------+---------+--------+------------+--------+
| emp_id | emp_name | dept_id | salary | hire_date  | city   |
+--------+----------+---------+--------+------------+--------+
Ask ChatGPT |        1 |  50000 | 2022-01-10 | Delhi  |
|    103 | Neha     |        1 |  55000 | 2020-07-20 | Delhi  |
|    107 | Pooja    |        1 |  48000 | 2022-09-30 | Delhi  |
|    110 | Vikas    |        1 |  51000 | 2023-01-18 | Delhi  |
+--------+----------+---------+--------+------------+--------+
4 rows in set (0.00 sec)
```

4. List employees from Delhi

```
SELECT * FROM employees WHERE city = 'Delhi';
```

```
+--------+----------+---------+--------+------------+--------+
| emp_id | emp_name | dept_id | salary | hire_date  | city   |
+--------+----------+---------+--------+------------+--------+
|    101 | Anuj     |        1 |  50000 | 2022-01-10 | Delhi  |
|    103 | Neha     |        1 |  55000 | 2020-07-20 | Delhi  |
|    107 | Pooja    |        1 |  48000 | 2022-09-30 | Delhi  |
|    110 | Vikas    |        1 |  51000 | 2023-01-18 | Delhi  |
+--------+----------+---------+--------+------------+--------+
4 rows in set (0.00 sec)
```

5. Show employees with salary greater than 60000.

```
SELECT emp name, salary FROM employees WHERE salary > 60000;
```

```
+----------+--------+
| emp_name | salary |
+----------+--------+
| Amit     |  70000 |
| Sneha    |  65000 |
| Karan    |  72000 |
| Meena    |  68000 |
+----------+--------+
4 rows in set (0.00 sec)
```

6. Find employees hired after 2021.

```
SELECT * FROM employees WHERE hire date > '2021-01-01';
```

```
+--------+----------+---------+--------+------------+--------+
| emp_id | emp_name | dept_id | salary | hire_date  | city   |
+--------+----------+---------+--------+------------+--------+
|    101 | Anuj     |       1 |  50000 | 2022-01-10 | Delhi  |
|    105 | Sneha    |       2 |  65000 | 2023-05-12 | Mumbai |
|    107 | Pooja    |       1 |  48000 | 2022-09-30 | Delhi  |
|    108 | Raj      |       2 |  52000 | 2021-12-11 | Chennai|
|    110 | Vikas    |       1 |  51000 | 2023-01-18 | Delhi  |
+--------+----------+---------+--------+------------+--------+
5 rows in set (0.00 sec)
```

7. Display unique cities.

```
SELECT DISTINCT city FROM employees;
```

```
+-----------+
| city      |
+-----------+
| Delhi     |
| Mumbai    |
| Pune      |
| Bangalore |
| Chennai   |
+-----------+
5 rows in set (0.00 sec)
```

8. Sort employees by salary (descending).

```
SELECT emp name, salary FROM employees ORDER BY salary DESC;
```

```
+----------+--------+
| emp_name | salary |
+----------+--------+
| Karan    |  72000 |
| Amit     |  70000 |
| Meena    |  68000 |
| Sneha    |  65000 |
| Ravi     |  60000 |
| Neha     |  55000 |
| Raj      |  52000 |
| Vikas    |  51000 |
| Anuj     |  50000 |
| Pooja    |  48000 |
+----------+--------+
10 rows in set (0.00 sec)
```

9. Count total employees.

```
SELECT COUNT(*) FROM employees;
```

```
+----------+
| COUNT(*) |
+----------+
|       10 |
+----------+
1 row in set (0.00 sec)
```

10. Find average salary.

```
SELECT AVG(salary) FROM employees;
```

```
+--------------+
| AVG(salary)  |
+--------------+
|      61100.0 |
+--------------+
1 row in set (0.00 sec)
```

Intermediate Level (11–20)

11. Find total salary paid per department.

```
SELECT dept id, SUM(salary) FROM employees GROUP BY dept id;
```

```
+---------+-------------+
| dept_id | SUM(salary) |
+---------+-------------+
|       1 |      204000 |
|       2 |      177000 |
|       3 |      210000 |
+---------+-------------+
3 rows in set (0.00 sec)
```

12. Count employees in each department.

```
SELECT dept id, SUM(salary) FROM employees GROUP BY dept id;
```

```
+---------+----------+
| dept_id | COUNT(*) |
+---------+----------+
|       1 |        4 |
|       2 |        3 |
|       3 |        3 |
+---------+----------+
3 rows in set (0.00 sec)
```

13. Find highest salary in each department.

```
SELECT dept id, MAX(salary) FROM employees GROUP BY dept id;
```

```
+---------+------------+
| dept_id | MAX(salary) |
+---------+------------+
|       1 |      55000 |
|       2 |      65000 |
|       3 |      72000 |
+---------+------------+
3 rows in set (0.00 sec)
```

14. Display employees along with department names.

```
SELECT e.emp_name, d.dept_name
      FROM employees e
      JOIN departments d
     ON e.dept id = d.dept id;
```

```
+----------+-----------+
| emp_name | dept_name |
+----------+-----------+
| Anuj     | HR        |
| Ravi     | Sales     |
| Neha     | HR        |
| Amit     | IT        |
| Sneha    | Sales     |
| Karan    | IT        |
| Pooja    | HR        |
| Raj      | Sales     |
| Meena    | IT        |
| Vikas    | HR        |
+----------+-----------+
10 rows in set (0.00 sec)
```

15. List employees working on Sales department.

```
SELECT e.emp name FROM employees e JOIN departments d ON e.dept id = d.dept id WHERE d.dept name = 'Sales';
```

```
+----------+
| emp_name |
+----------+
| Ravi     |
| Sneha    |
| Raj      |
+----------+
3 rows in set (0.00 sec)
```

16. Find employees whose name starts with 'A'.

```
SELECT * FROM employees WHERE emp name LIKE 'A%';
```

```
+--------+----------+---------+--------+------------+-------+
| emp_id | emp_name | dept_id | salary | hire_date  | city  |
+--------+----------+---------+--------+------------+-------+
|    101 | Anuj     |       1 |  50000 | 2022-01-10 | Delhi |
|    104 | Amit     |       3 |  70000 | 2019-11-01 | Pune  |
+--------+----------+---------+--------+------------+-------+
2 rows in set (0.00 sec)
```

17. Get employees with salary between 50000 and 70000.

```
SELECT * FROM employees WHERE salary BETWEEN 50000 AND 70000;
```

```
+--------+----------+---------+--------+------------+-----------+
| emp_id | emp_name | dept_id | salary | hire_date  | city      |
+--------+----------+---------+--------+------------+-----------+
|    101 | Anuj     |       1 |  50000 | 2022-01-10 | Delhi     |
|    102 | Ravi     |       2 |  60000 | 2021-03-15 | Mumbai    |
|    103 | Neha     |       1 |  55000 | 2020-07-20 | Delhi     |
|    104 | Amit     |       3 |  70000 | 2019-11-01 | Pune      |
|    105 | Sneha    |       2 |  65000 | 2023-05-12 | Mumbai    |
|    108 | Raj      |       2 |  52000 | 2021-12-11 | Chennai   |
|    109 | Meena    |       3 |  68000 | 2020-02-14 | Pune      |
|    110 | Vikas    |       1 |  51000 | 2023-01-18 | Delhi     |
+--------+----------+---------+--------+------------+-----------+
8 rows in set (0.00 sec)
```

18. Find employees not from Mumbai.

```
SELECT * FROM employees WHERE city != 'Mumbai';
```

```
+--------+----------+---------+--------+------------+-----------+
| emp_id | emp_name | dept_id | salary | hire_date  | city      |
+--------+----------+---------+--------+------------+-----------+
|    101 | Anuj     |       1 |  50000 | 2022-01-10 | Delhi     |
|    103 | Neha     |       1 |  55000 | 2020-07-20 | Delhi     |
|    104 | Amit     |       3 |  70000 | 2019-11-01 | Pune      |
|    106 | Karan    |       3 |  72000 | 2018-08-25 | Bangalore |
|    107 | Pooja    |       1 |  48000 | 2022-09-30 | Delhi     |
|    108 | Raj      |       2 |  52000 | 2021-12-11 | Chennai   |
|    109 | Meena    |       3 |  68000 | 2020-02-14 | Pune      |
|    110 | Vikas    |       1 |  51000 | 2023-01-18 | Delhi     |
+--------+----------+---------+--------+------------+-----------+
8 rows in set (0.00 sec)
```

19. Show employees hired in 2022.

```
SELECT * FROM employees WHERE YEAR(hire date) = 2022;
```

```
+--------+----------+---------+--------+------------+--------+
| emp_id | emp_name | dept_id | salary | hire_date  | city   |
+--------+----------+---------+--------+------------+--------+
|    101 | Anuj     |       1 |  50000 | 2022-01-10 | Delhi  |
|    107 | Pooja    |       1 |  48000 | 2022-09-30 | Delhi  |
+--------+----------+---------+--------+------------+--------+
2 rows in set (0.00 sec)
```

20. Find top 3 highest paid employees.

```
SELECT * FROM employees ORDER BY salary DESC LIMIT 3;
```

```
+--------+----------+---------+--------+------------+-----------+
| emp_id | emp_name | dept_id | salary | hire_date  | city      |
+--------+----------+---------+--------+------------+-----------+
|    106 | Karan    |       3 |  72000 | 2018-08-25 | Bangalore |
|    104 | Amit     |       3 |  70000 | 2019-11-01 | Pune      |
|    109 | Meena    |       3 |  68000 | 2020-02-14 | Pune      |
+--------+----------+---------+--------+------------+-----------+
3 rows in set (0.00 sec)
```

Joins (21–25)

21. INNER JOIN Employees and Departments.

```
SELECT e.emp_name, d.dept_name
FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id;
```

```
+----------+-----------+
| emp_name | dept_name |
+----------+-----------+
| Anuj     | HR        |
| Ravi     | Sales     |
| Neha     | HR        |
| Amit     | IT        |
| Sneha    | Sales     |
| Karan    | IT        |
| Pooja    | HR        |
| Raj      | Sales     |
| Meena    | IT        |
| Vikas    | HR        |
+----------+-----------+
10 rows in set (0.00 sec)
```

22. LEFT JOIN to show all employees with department names.

```
SELECT e.emp_name, d.dept_name
FROM employees e
LEFT JOIN departments d
ON e.dept id = d.dept id;
```

```
+----------+-----------+
| emp_name | dept_name |
+----------+-----------+
| Anuj     | HR        |
| Ravi     | Sales     |
| Neha     | HR        |
| Amit     | IT        |
| Sneha    | Sales     |
| Karan    | IT        |
| Pooja    | HR        |
| Raj      | Sales     |
| Meena    | IT        |
| Vikas    | HR        |
+----------+-----------+
10 rows in set (0.00 sec)
```

23. Find employees working in IT department.

```
SELECT e.emp_name
FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id
WHERE d.dept name = 'IT';
```

```
+----------+
| emp_name |
+----------+
| Amit     |
| Karan    |
| Meena    |
+----------+
3 rows in set (0.00 sec)
```

24. Show project names with department names.

```
SELECT p.project_name, d.dept_name
FROM projects p
JOIN departments d
ON p.dept id = d.dept id;
```

```
+--------------+-----------+
| project_name | dept_name |
+--------------+-----------+
| Alpha        | HR        |
| Beta         | Sales     |
| Gamma        | IT        |
| Delta        | Sales     |
+--------------+-----------+
4 rows in set (0.00 sec)
```

25. Find employees and their project budgets (via department).

```
SELECT e.emp_name, p.budget
FROM employees e
JOIN projects p
ON e.dept_id = p.dept_id;
```

```
+----------+--------+
| emp_name | budget |
+----------+--------+
| Anuj     | 100000 |
| Neha     | 100000 |
| Pooja    | 100000 |
| Vikas    | 100000 |
| Ravi     | 150000 |
| Sneha    | 150000 |
| Raj      | 150000 |
| Ravi     | 120000 |
| Sneha    | 120000 |
| Raj      | 120000 |
| Amit     | 200000 |
| Karan    | 200000 |
| Meena    | 200000 |
+----------+--------+
13 rows in set (0.00 sec)
```

Advanced Level (26–30)

26. Find second highest salary.

```
SELECT MAX(salary)
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

```
+-------------+
| MAX(salary) |
+-------------+
|       70000 |
+-------------+
1 row in set (0.00 sec)
```

27. Rank employees by salary.

```
SELECT emp_name, salary,
RANK() OVER (ORDER BY salary DESC) AS rnk
FROM employees;
```

```
+----------+--------+-----+
| emp_name | salary | rnk |
+----------+--------+-----+
| Karan    |  72000 |   1 |
| Amit     |  70000 |   2 |
| Meena    |  68000 |   3 |
| Sneha    |  65000 |   4 |
| Ravi     |  60000 |   5 |
| Neha     |  55000 |   6 |
| Raj      |  52000 |   7 |
| Vikas    |  51000 |   8 |
| Anuj     |  50000 |   9 |
| Pooja    |  48000 |  10 |
+----------+--------+-----+
10 rows in set (0.00 sec)
```

28. Find employees earning above average salary.

```
SELECT *
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees);
```

emp_id	emp_name	dept_id	salary	hire_date	city
104	Amit	3	70000	2019-11-01	Pune
105	Sneha	2	65000	2023-05-12	Mumbai
106	Karan	3	72000	2018-08-25	Bangalore
109	Meena	3	68000	2020-02-14	Pune

4 rows in set (0.00 sec)

29. Get department with highest total salary.

```
SELECT dept_id, SUM(salary) AS total
FROM employees
GROUP BY dept_id
ORDER BY total DESC
LIMIT 1;
```

dept_id	total
3	210000

1 row in set (0.00 sec)

30. Find employees who were absent on 2024-03-01.

```sql
SELECT e.emp_name
FROM employees e
JOIN attendance a
ON e.emp_id = a.emp_id
WHERE a.status = 'Absent';
```

```
+----------+
| emp_name |
+----------+
| Ravi     |
| Sneha    |
| Meena    |
+----------+
3 rows in set (0.00 sec)
```